Lab03 - ME315

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```r
library(readr)
library(dplyr)
```

```
Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

    filter, lag

Os seguintes objetos são mascarados por 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(tidyr)
library(stringr)
library(leaflet)
library(webshot2)
```

Questão 1

```r
airports <- read_csv("airports.csv",
                     col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE))
```

```
Rows: 322 Columns: 5
-- Column specification --------------------------------------------------------
Delimiter: ","
```

```
chr (3): IATA_CODE, CITY, STATE
dbl (2): LATITUDE, LONGITUDE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
airlines <- read_csv("airlines.csv")
```

```
Rows: 14 Columns: 2
-- Column specification --------------------------------------------------------
Delimiter: ","
chr (2): IATA_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
processa_bloco <- function(bloco, pos) {
  bloco %>%
    filter(!str_starts(DESTINATION_AIRPORT, "1")) %>%
    drop_na() %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarise(soma_atraso = sum(ARRIVAL_DELAY), n_voos = n(), .groups = "drop")
}

estatisticas_parciais <- read_csv_chunked(
  "flights.csv.zip",
  callback = DataFrameCallback$new(processa_bloco),
  chunk_size = 1e6,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)

flights <- estatisticas_parciais %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    atraso_medio = sum(soma_atraso) / sum(n_voos),
    n_voos = sum(n_voos),
    .groups = "drop"
  )
```

Questão 2

```r
# A chave na tabela 'flights' é 'DESTINATION_AIRPORT'
# A chave na tabela 'airports' é 'IATA_CODE'

anti_join(flights, airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
```

```
# A tibble: 0 x 3
# i 3 variables: DESTINATION_AIRPORT <chr>, atraso_medio <dbl>, n_voos <int>
```

```r
flights <- flights %>%
  left_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
```

Questão 3

```r
airports %>%
  count(STATE, name = "n_aeroportos") %>%
  arrange(desc(n_aeroportos))
```

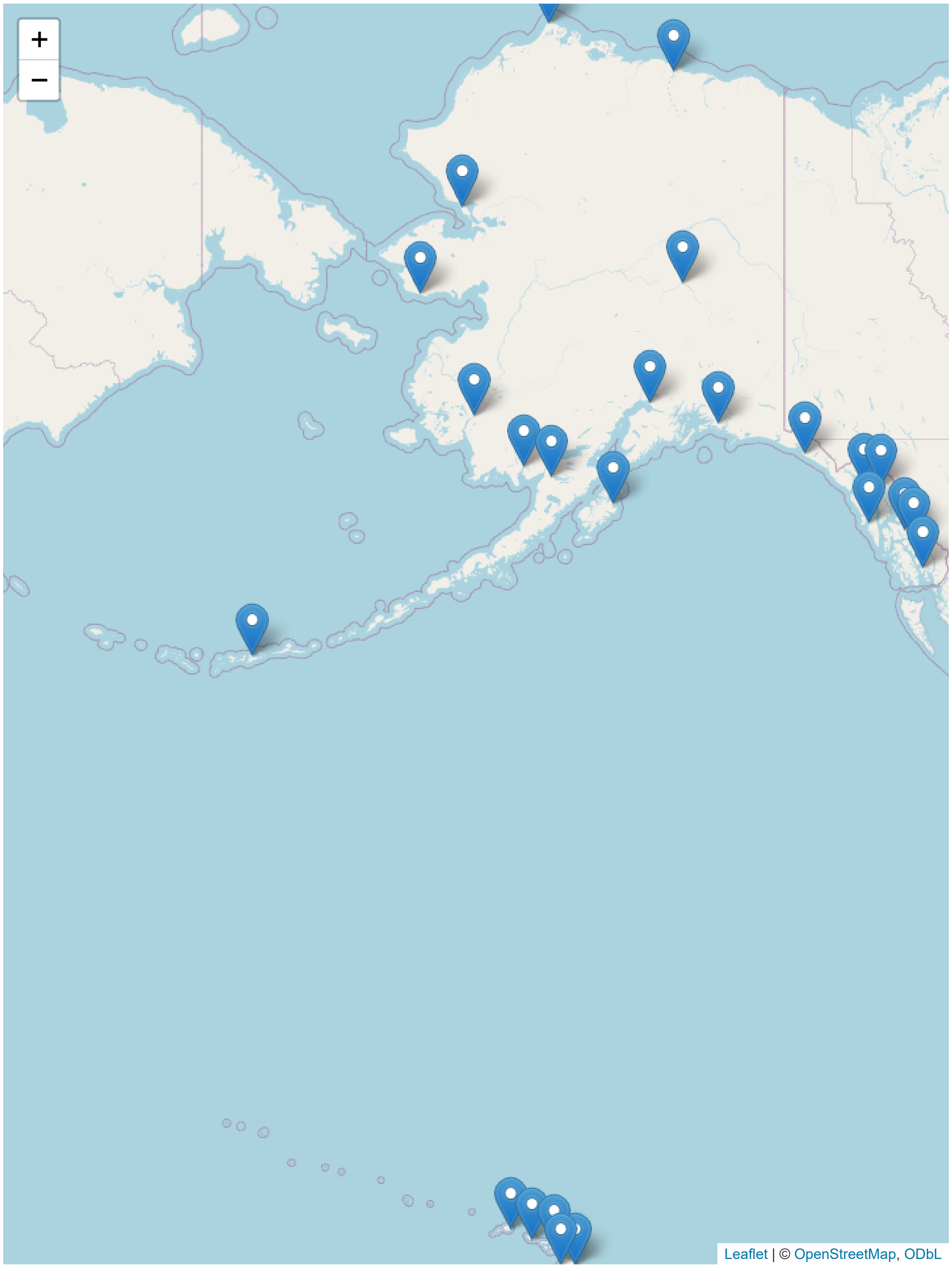
```
# A tibble: 54 x 2
   STATE n_aeroportos
   <chr>        <int>
 1 TX              24
 2 CA              22
 3 AK              19
 4 FL              17
 5 MI              15
 6 NY              14
 7 CO              10
 8 MN               8
 9 MT               8
10 NC               8
# i 44 more rows
```

Questão 4

```
dados_mapa <- flights %>%
  drop_na(LATITUDE, LONGITUDE)

this <- leaflet(data = dados_mapa) %>%
  addTiles() %>%
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      DESTINATION_AIRPORT, " - ", CITY, ", ", STATE, "<br>",
      "Atraso médio: ", round(atraso_medio, 1), " min<br>",
      "Voos: ", n_voos
    )
  )

this
```


+
−
Leaflet | © OpenStreetMap, ODbL

```r
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-31 19:10:35 -03"
```